

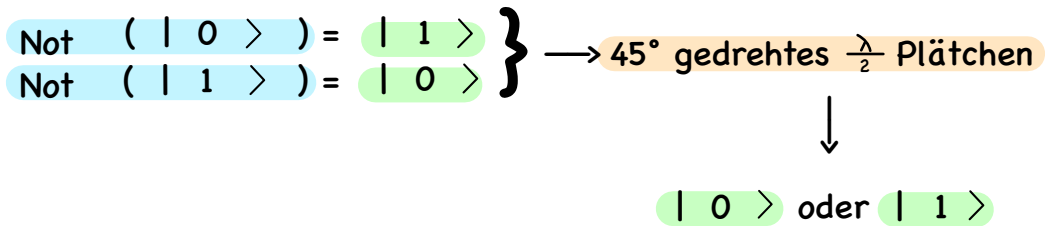
Quantencomputer Gatter

Made by

Jannes Ruder



Not

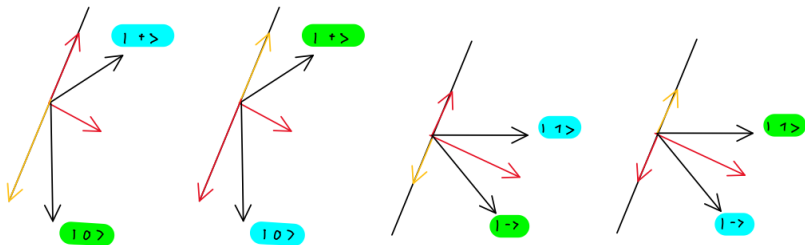


Hadarmad

$$\begin{aligned}
 \text{Hadarmad } (|0\rangle) &= |+\rangle \\
 \text{Hadarmad } (|1\rangle) &= |-\rangle \\
 \text{Hadarmad } (|+\rangle) &= |0\rangle \\
 \text{Hadarmad } (|-\rangle) &= |1\rangle
 \end{aligned}
 \left. \vphantom{\begin{aligned} \text{Hadarmad } (|0\rangle) &= |+\rangle \\ \text{Hadarmad } (|1\rangle) &= |-\rangle \\ \text{Hadarmad } (|+\rangle) &= |0\rangle \\ \text{Hadarmad } (|-\rangle) &= |1\rangle \end{aligned}} \right\} \rightarrow 22,5^\circ \text{ gedrehtes } \frac{\lambda}{2} \text{ Plättchen}$$

↓

$|0\rangle$, $|1\rangle$, $|+\rangle$ oder $|-\rangle$



C-Not

$$\text{C-Not} (| 0 \rangle | 0 \rangle) = | 0 \rangle | 0 \rangle$$

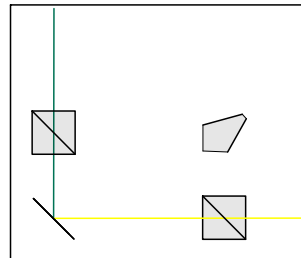
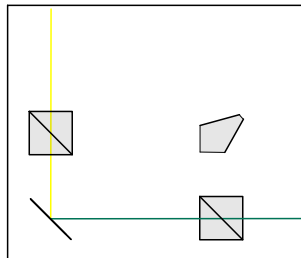
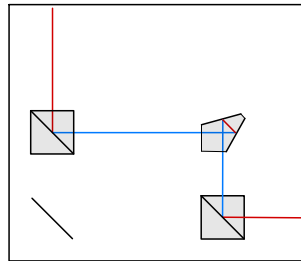
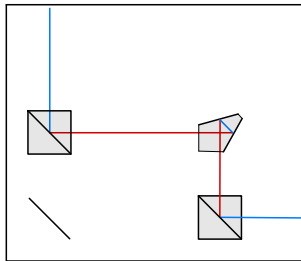
$$\text{C-Not} (| 0 \rangle | 1 \rangle) = | 0 \rangle | 1 \rangle$$

$$\text{C-Not} (| 1 \rangle | 0 \rangle) = | 1 \rangle | 1 \rangle$$

$$\text{C-Not} (| 1 \rangle | 1 \rangle) = | 1 \rangle | 0 \rangle$$



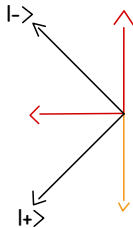
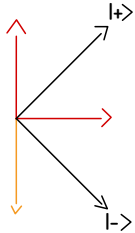
$| 0 \rangle | 0 \rangle$ — blue
 $| 0 \rangle | 1 \rangle$ — red
 $| 1 \rangle | 0 \rangle$ — yellow
 $| 1 \rangle | 1 \rangle$ — green
 Pentaprisma
 Beamsplitter
 Spiegel



$| 0 \rangle$ oder $| 1 \rangle$

Z

$$\begin{aligned} Z (| + \rangle) &= | - \rangle \\ Z (| - \rangle) &= | + \rangle \end{aligned} \quad \left. \vphantom{\begin{aligned} Z (| + \rangle) &= | - \rangle \\ Z (| - \rangle) &= | + \rangle \end{aligned}} \right\} \longrightarrow 0^\circ \text{ gedrehtes } \frac{\lambda}{2} \text{ Plättchen}$$



$| + \rangle$ oder $| - \rangle$

T

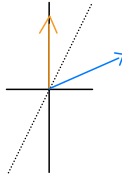
$$T(|0\rangle) = |0\rangle$$

$$T(|1\rangle) = |1\rangle$$

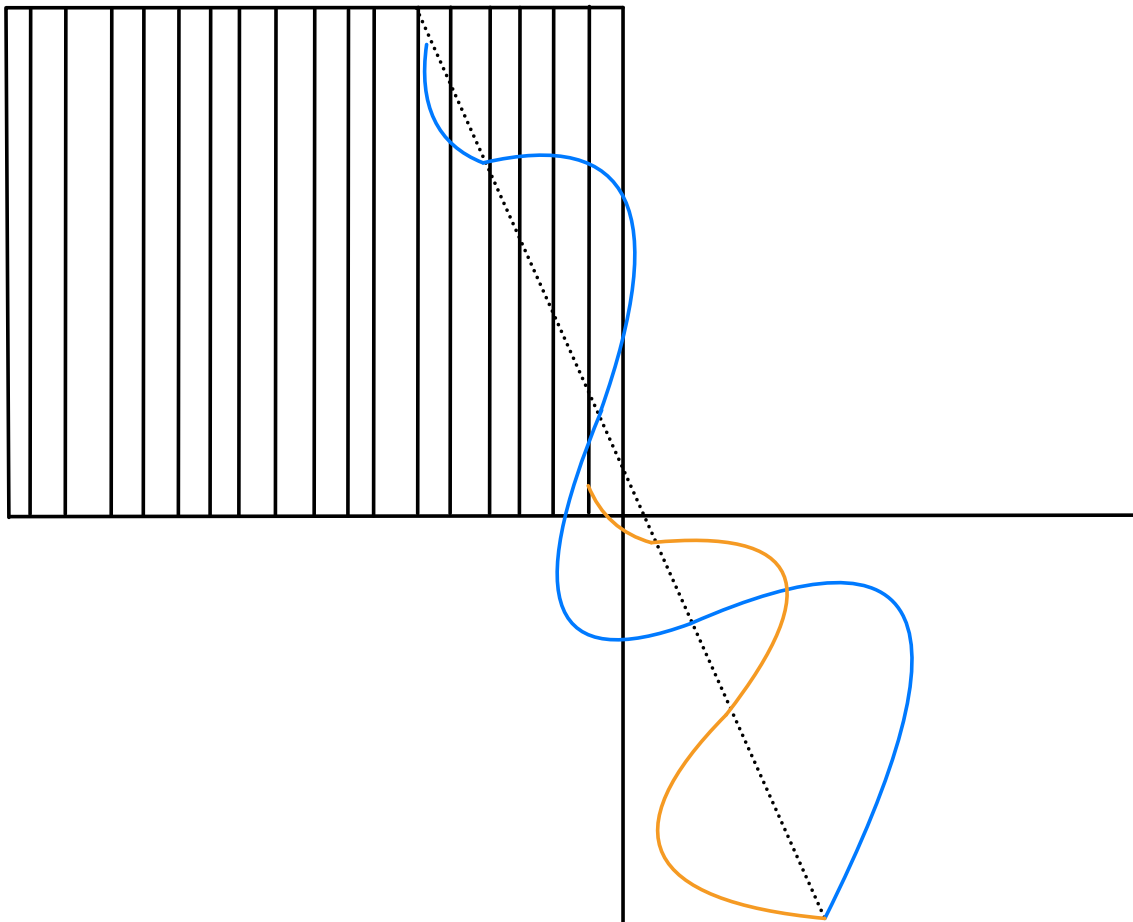
$$T(|+\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$T(|-\rangle) = |0\rangle - |1\rangle$$

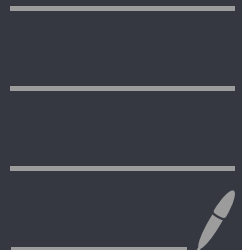
}



Polarisatoren



Quantencomputer Berechnungen und Formeln



Vektor Schreibweise für Qubits

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Kontroll Qubit Ziel Qubit

$$|1\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|0\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Matrix Schreibweise für Gatter

$$\text{C-Not} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\text{Hadamard} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\text{Not} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\text{T} = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$$

$$\text{Beamsplitter} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Spiegel} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Beispiel Berechnungen des C-Not

$$|1\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |1\rangle \otimes |0\rangle$$

$$|1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |1\rangle \otimes |1\rangle$$

$$|0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |0\rangle \otimes |0\rangle$$

$$|0\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |0\rangle \otimes |1\rangle$$

Beispiel Berechnungen des Hadarmad Gatters

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 0 + -1 \cdot 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = | - \rangle$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 \\ 1 \cdot 1 + -1 \cdot 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = | + \rangle$$

Beispiel Berechnungen des NOT Gatters

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{bmatrix} 0 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 0 + 0 \cdot 1 \end{bmatrix} = | 0 \rangle$$

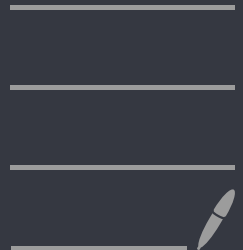
$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 \end{bmatrix} = | 1 \rangle$$

Beispiel Berechnungen des Z Gatters

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + -1 \cdot 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = | - \rangle$$

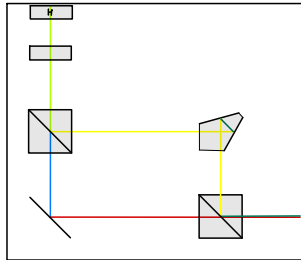
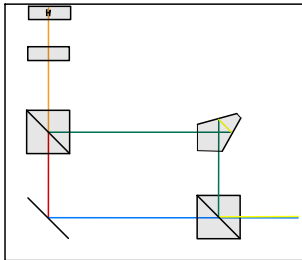
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot -1 + -1 \cdot -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = | + \rangle$$

Quantencomputer Algorithmen



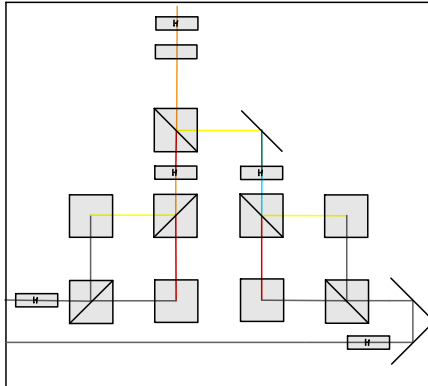
Deutsch - Algorithmus

$|0\rangle|0\rangle$ — blue
 $|0\rangle|1\rangle$ — red
 $|1\rangle|0\rangle$ — yellow
 $|1\rangle|1\rangle$ — green
 $|+\rangle|0\rangle$ — light green
 $|+\rangle|1\rangle$ — orange
 Pentaprisma 
 Beamsplitter 
 Spiegel 
 Hologramm 
 Hadarmad 



Deutsch-Jozsa- Algorithmus

$|0\rangle |0\rangle$ — blue
 $|0\rangle |1\rangle$ — red
 $|1\rangle |0\rangle$ — yellow
 $|1\rangle |1\rangle$ — green
 $|+\rangle |0\rangle$ — light green
 $|+\rangle |1\rangle$ — orange
 $|-\rangle |0\rangle$ — purple
 $|-\rangle |1\rangle$ — cyan
 Beamsplitter 
 Spiegel 
 Hologramm 
 Pentaprisma 
 Hadarmad 

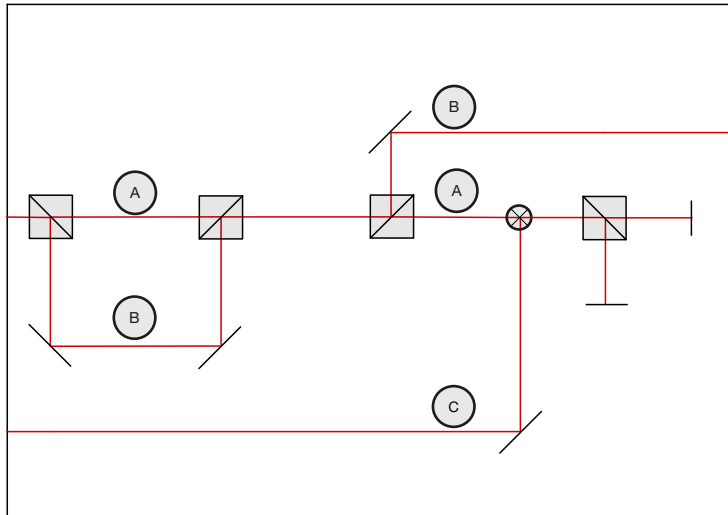


QPU



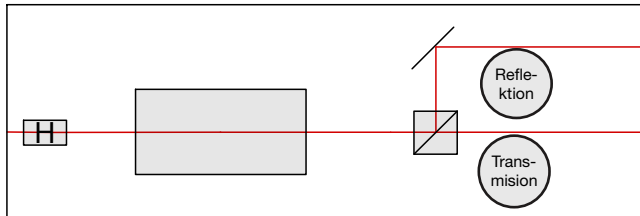
Quanten-Teleportation

Beamsplitter 
Spiegel 
Fasserkopler 



Quanten-Teleportation

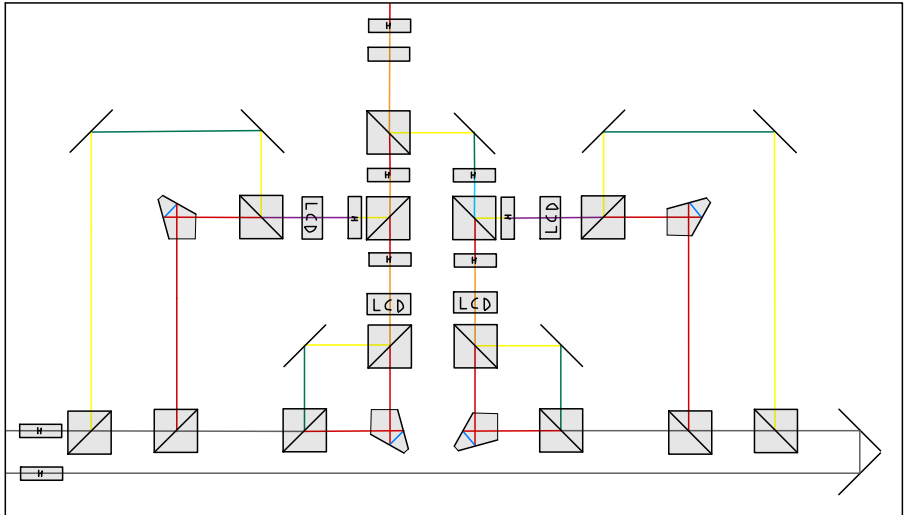
Beamsplitter
Spiegel
PolaRITE3
Hadarmad



Deutsch-Josza-Algorithmus

$|0\rangle |0\rangle$ — blue
 $|0\rangle |1\rangle$ — red
 $|1\rangle |0\rangle$ — yellow
 $|1\rangle |1\rangle$ — green
 $|+\rangle |0\rangle$ — light green
 $|+\rangle |1\rangle$ — orange
 $|-\rangle |0\rangle$ — purple
 $|-\rangle |1\rangle$ — cyan

Beamsplitter 
 Spiegel 
 Hologramm 
 Pentaprisma 
 Hadarmad 
 LCD Display 



Deutsch-Josza-Algorithmus

$|0\rangle |0\rangle$ — blue
 $|0\rangle |1\rangle$ — red
 $|1\rangle |0\rangle$ — yellow
 $|1\rangle |1\rangle$ — green
 $|+\rangle |0\rangle$ — light green
 $|+\rangle |1\rangle$ — orange
 $|-\rangle |0\rangle$ — purple
 $|-\rangle |1\rangle$ — cyan

Beamsplitter 
 Spiegel 
 Hologramm 
 Pentaprisma 
 Hadarmad 
 LCD Display 

